

# **Advanced Spatio Temporal Demand Forecasting and Adaptive Seat Allocation using AI**

R26-IT-127

Project Proposal Report

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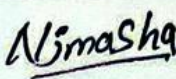
Sri Lanka Institute of Information Technology Sri Lanka

March 2026

## DECLARATION

### DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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## ABSTRACT

Urban railway transportation systems face increasing challenges in managing fluctuating passenger demand and ensuring efficient seat utilization. Traditional planning approaches rely on static scheduling and historical averages which fail to capture complex variations in passenger demand across time and locations. As a result railway systems frequently experience overcrowding during peak periods and underutilization during off-peak hours leading to reduced passenger comfort inefficient resource usage and increased operational costs.

Despite advancements in transportation demand forecasting many existing approaches primarily focus on temporal prediction using statistical and machine learning models. These methods often fail to capture spatial dependencies between stations and do not integrate demand predictions with operational decision-making processes such as seat allocation. Furthermore most railway systems continue to use static or rule-based seat allocation strategies which do not adapt dynamically to changing demand patterns.

To address these limitations this research proposes an AI-driven spatio-temporal demand forecasting and adaptive seat allocation framework for railway transportation systems. The proposed solution integrates deep learning models such as Long Short-Term Memory networks for temporal modeling and Graph Neural Networks for capturing spatial relationships within railway networks. Based on predicted demand patterns an adaptive seat allocation model is developed using optimization techniques to dynamically allocate seats across routes and time periods.

The system is implemented using modern artificial intelligence technologies including Python TensorFlow or PyTorch and data analytics frameworks. The expected outcomes include improved demand prediction accuracy optimized seat utilization reduced overcrowding and enhanced passenger experience. In addition the proposed framework supports data-driven decision-making for railway operators and improves operational efficiency.

From a social perspective the system enhances accessibility comfort and reliability of public transportation for daily commuters. From a commercial perspective it enables better resource utilization reduces operational costs and improves revenue management for railway operators. The research also contributes to sustainable transportation development by promoting efficient use of existing infrastructure.

**Keywords:** Spatio-Temporal Forecasting Artificial Intelligence Railway Demand Prediction Adaptive Seat Allocation Smart Transportation.

## ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor Ms. Chathurangika Kahadawaarachchi for her valuable guidance continuous support and constructive feedback throughout the development of this research proposal. Her expertise and insights have greatly contributed to shaping the direction and quality of this research.

I would also like to thank the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment learning resources and research support necessary for the successful preparation of this proposal.

Finally I would like to extend my heartfelt appreciation to my family and friends for their encouragement patience and continuous support throughout the preparation of this research proposal.

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## LIST OF ABBREVIATIONS

Table 1: Abbreviations

|       |                                           |
|-------|-------------------------------------------|
| AI    | Artificial Intelligence                   |
| ARIMA | Auto-Regressive Integrated Moving Average |
| GNN   | Graph Neural Network                      |
| GRU   | Gated Recurrent Unit                      |
| LSTM  | Long Short-Term Memory                    |
| MAE   | Mean Absolute Error                       |
| MAPE  | Mean Absolute Percentage Error            |
| MILP  | Mixed Integer Linear Programming          |
| ML    | Machine Learning                          |
| OD    | Origin–Destination                        |
| RMSE  | Root Mean Square Error                    |
| TCN   | Temporal Convolutional Network            |

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# 1. INTRODUCTION

## 1.1 Background and Context

Transportation systems play a crucial role in modern urban development by enabling mobility supporting economic activities and improving the overall quality of life. Among various modes of public transport railway systems are widely used due to their ability to efficiently transport large numbers of passengers at relatively low cost. However with rapid urbanization and population growth railway transportation systems are increasingly facing challenges related to passenger demand management and resource utilization.

Globally urban populations are expected to reach approximately 68% of the total population by 2050 increasing the demand for public transportation services. As a result railway systems in many countries experience severe congestion during peak hours while operating below capacity during off-peak periods. In Sri Lanka railway transportation remains one of the most commonly used modes of travel particularly for daily commuting. According to national transportation reports thousands of passengers rely on trains daily leading to overcrowding during peak times especially on major routes such as Colombo-based railway lines. This imbalance results in reduced passenger comfort increased waiting times and inefficient use of available seating capacity.

The core real-world problem lies in the inability of existing railway systems to accurately predict passenger demand and dynamically allocate resources based on those predictions. Most railway operators rely on static scheduling fixed seat allocation policies and historical averages to manage capacity. However passenger demand is highly dynamic and influenced by multiple factors including geographical location weather conditions public events and passenger travel behavior. These factors generate complex spatio-temporal demand patterns that cannot be effectively captured using traditional planning approaches.

This issue affects multiple stakeholders. Passengers experience overcrowding discomfort and uncertainty in travel while railway operators face challenges in optimizing resource utilization and maintaining service efficiency. Additionally inefficient transportation systems contribute to increased operational costs and reduced sustainability impacting the broader goal of developing smart and sustainable cities.

To address demand prediction challenges various research approaches and systems have been developed. Traditional statistical models such as ARIMA and modern machine learning techniques such as Support Vector Machines and Random Forests have been used to forecast passenger demand. More recently deep learning models including Long Short-Term Memory networks and Graph Neural Networks have been introduced to capture complex temporal and spatial relationships in transportation data. While these models have significantly improved forecasting accuracy they are often used only for prediction purposes and are not integrated into operational decision-making processes.

Similarly existing railway seat allocation systems rely on static rules predefined quotas or basic optimization techniques. Although some advanced methods such as linear programming and heuristic optimization have been applied these approaches do not adapt dynamically to real-time or predicted demand fluctuations. As a result there is a disconnect between demand forecasting and resource allocation leading to inefficient capacity utilization.

To overcome these limitations this research proposes an integrated AI-based framework that combines advanced spatio-temporal demand forecasting with adaptive seat allocation for railway transportation systems. The proposed solution consists of two main components: a spatio-temporal demand forecasting module that utilizes deep learning and graph-based models to predict passenger demand across railway stations and an adaptive seat allocation module that dynamically allocates seating capacity based on predicted demand using optimization techniques. By integrating these components the system aims to improve prediction accuracy and enable real-time data-driven decision-making for railway operations.

This research aligns with the **United Nations Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities** which emphasizes the development of safe efficient and sustainable transportation systems. By improving railway capacity utilization and reducing overcrowding the proposed solution contributes to more sustainable urban mobility and better passenger experience.

Furthermore, this study falls within the research cluster of **Artificial Intelligence and Data Science for Smart Transportation Systems** focusing on the application of AI machine learning and optimization techniques to solve real-world transportation challenges. The integration of spatiotemporal modeling and adaptive decision-making mechanisms represents a significant step toward intelligent transportation systems and smart city development.

## 1.2 Literature Review

Transportation demand forecasting has become a critical research area in intelligent transportation systems, particularly in railway transportation, due to increasing urbanization and the rapid growth of passenger mobility. Accurate prediction of passenger demand is essential for railway operators to optimize seat allocation, reduce overcrowding, and improve service efficiency. In recent years, artificial intelligence (AI) techniques have significantly enhanced demand forecasting by enabling the modeling of complex spatio-temporal patterns that capture both temporal trends and spatial interactions across railway networks.

Early research in railway demand forecasting primarily relied on statistical time-series models such as the Auto-Regressive Integrated Moving Average (ARIMA). These models use historical passenger data to predict future demand by capturing linear trends and seasonal variations. According to Box et al. [1], ARIMA models perform well for structured datasets with stable temporal patterns. However, railway demand is highly dynamic and influenced by multiple factors such as station connectivity, peak-hour

variations, and passenger behavior. As a result, ARIMA models struggle to capture nonlinear relationships and fail to incorporate spatial dependencies between different railway stations.

To address these limitations, machine learning approaches such as Support Vector Machines (SVM), Random Forests, and Gradient Boosting have been introduced. These models are capable of handling nonlinear data and large-scale transportation datasets. Breiman [2] demonstrated that Random Forest algorithms improve prediction accuracy by aggregating multiple decision trees, while Drucker et al. [3] showed that SVM-based regression models perform effectively in demand prediction tasks. Despite these advantages, machine learning models are limited in capturing temporal dependencies in sequential data, which is a key requirement for forecasting passenger demand over time.

With the advancement of deep learning, neural network-based models have gained significant attention in railway demand forecasting. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are widely used due to their ability to model sequential data and capture long-term temporal dependencies. Hochreiter and Schmidhuber [4] introduced LSTM networks to address the limitations of traditional RNNs. Studies such as Ma et al. [5] demonstrated that LSTM models significantly outperform traditional statistical and machine learning approaches in traffic and passenger demand prediction tasks by effectively learning complex temporal patterns.

However, railway transportation systems are inherently spatial, where demand at one station is influenced by neighboring stations and route connectivity. LSTM models, while effective for temporal modeling, do not explicitly capture spatial relationships. To address this limitation, Graph Neural Networks (GNNs) have been introduced. Kipf and Welling [6] proposed Graph Convolutional Networks (GCNs), which enable learning from graph-structured data where nodes represent stations and edges represent railway connections.

Recent research has focused on integrating spatial and temporal modeling techniques to develop spatiotemporal forecasting models. For example, Li et al. [7] proposed the Diffusion Convolutional Recurrent Neural Network (DCRNN), which combines graph convolution with recurrent neural networks to capture both spatial and temporal dependencies. Similarly, Yu et al. [8] introduced the Spatio-Temporal Graph Convolutional Network (STGCN), which applies convolution operations across both spatial and temporal dimensions. These models have demonstrated superior performance in predicting transportation demand by effectively capturing dynamic interactions within railway networks.

More recently, attention-based and transformer architectures have been applied to spatio-temporal forecasting. Vaswani et al. [9] introduced the Transformer model, which has been adapted for timeseries forecasting to capture long-range dependencies. In railway systems, attention-based models dynamically assign importance to different stations and time intervals, improving prediction accuracy.

Studies such as Zheng et al. [10] highlight that combining graph neural networks with attention mechanisms further enhances the performance of large-scale transportation forecasting models.

In real-world railway systems, demand forecasting and seat allocation are typically handled using separate modules. Many railway reservation systems rely on predefined quotas, historical booking patterns, and rule-based scheduling to allocate seats. These systems are widely used in national railway networks and are effective for basic reservation management.

In addition, intelligent transportation systems (ITS) have been implemented in some railway networks to monitor passenger flow and optimize scheduling. Data-driven approaches are used to analyze ticketing data, passenger counts, and travel patterns. However, these systems primarily focus on monitoring and reporting rather than predictive and adaptive decision-making.

Some modern transportation platforms incorporate predictive analytics to estimate demand. For example, large-scale transport systems use machine learning models to forecast passenger flow and adjust schedules accordingly. However, these forecasting systems often operate independently of seat allocation mechanisms. Seat allocation decisions are still performed using static rules or simple optimization techniques, without fully utilizing real-time demand predictions.

Despite significant advancements in both research and real-world applications, several limitations remain in current railway transportation systems. One of the major limitations is the separation between demand forecasting and seat allocation. Most research studies focus either on improving prediction accuracy or optimizing resource allocation independently. As highlighted by Yao et al. [11], this lack of integration prevents transportation systems from dynamically adapting to changing passenger demand, leading to inefficiencies such as overcrowding during peak hours and underutilized capacity during offpeak periods.

Another limitation is the inability of traditional models to capture both spatial and temporal dependencies simultaneously. While LSTM models effectively capture temporal patterns, they fail to consider spatial relationships between stations. Conversely, graph-based models capture spatial interactions but may not fully model temporal dynamics when used independently.

Additionally, many railway reservation systems rely on static scheduling and rule-based seat allocation methods. These approaches do not adapt to real-time demand fluctuations and fail to respond dynamically to unexpected changes in passenger demand. Although optimization techniques such as linear programming, Genetic Algorithms, and Particle Swarm Optimization have been applied to resource allocation problems, they often rely on predefined constraints and do not incorporate real-time demand predictions [12], [13].

Furthermore, advanced AI models require large volumes of high-quality data and significant computational resources, which may not be available in all railway systems, particularly in developing

regions. Another key challenge is the lack of integration between forecasting outputs and operational decision-making processes. As a result, even accurate demand predictions are not effectively utilized in improving seat allocation and scheduling decisions [14].

Based on the analysis of existing research and systems, it is evident that while significant progress has been made in spatio-temporal demand forecasting and optimization techniques, there is still a lack of integrated solutions that combine both aspects in railway transportation systems. Current systems fail to dynamically allocate seats based on predicted demand, limiting their operational efficiency.

Therefore, there is a clear need for an intelligent framework that integrates advanced spatio-temporal demand forecasting with adaptive seat allocation mechanisms. This research aims to address this gap by developing an AI-based system that predicts passenger demand across railway networks and dynamically allocates seats based on those predictions, thereby improving resource utilization and passenger experience.

### **1.3 Research Gap**

Although significant research has been conducted in transportation demand forecasting and railway capacity optimization, several critical limitations remain in existing studies.

First, many railway analytics systems primarily focus on descriptive analytics and operational monitoring rather than predictive and prescriptive decision-support frameworks. Recent studies on railway data analytics emphasize the need to transition from manual planning and heuristic-based operations toward intelligent, data-driven systems capable of supporting real-time decision making. However, in practice, many railway systems—particularly in developing regions such as Sri Lanka—still rely heavily on traditional planning approaches without integrating predictive analytics into operational workflows.

Second, classical time-series forecasting methods such as ARIMA and more recent scalable models such as Prophet provide strong baseline performance for temporal prediction. However, these models are inherently limited to temporal analysis and fail to capture nonlinear dynamics and spatial dependencies between stations. Since railway demand is influenced by network structure and interstation interactions, these models are insufficient for representing the complex nature of passenger mobility.

Third, deep learning models such as Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Temporal Convolutional Networks (TCN) have demonstrated improved performance in modeling nonlinear temporal patterns. Nevertheless, many of these approaches focus primarily on temporal sequences and do not fully exploit the graph-structured characteristics of transportation networks.

Railway systems consist of interconnected stations and origin–destination flows, and ignoring spatial relationships can significantly limit forecasting accuracy.

Recent advances in spatio-temporal graph learning models, including Diffusion Convolutional Recurrent Neural Networks (DCRNN), Spatio-Temporal Graph Convolutional Networks (STGCN), and Graph WaveNet, have shown promising results in capturing both temporal patterns and spatial dependencies. However, the application of these models to railway-specific problems such as origin–destination demand forecasting and capacity-aware seat allocation remains limited and underexplored.

Another major limitation in existing literature is the separation between demand forecasting and resource allocation. Most studies treat demand prediction and seat allocation as independent problems. While forecasting models estimate passenger demand, seat allocation systems typically rely on static rules or predefined scheduling mechanisms that do not adapt dynamically to predicted demand variations. This disconnect reduces the practical impact of forecasting models in real-world railway operations.

Furthermore, recent research on railway seat allocation has introduced fairness-aware optimization techniques, including Mixed Integer Linear Programming (MILP) and robust optimization approaches. While these models incorporate uncertainty and fairness constraints, they are rarely integrated with advanced demand forecasting models that provide real-time or predictive demand inputs.

In addition, existing studies often overlook the operational and data constraints present in developing railway systems such as those in Sri Lanka. Challenges such as limited availability of origin–destination data, privacy restrictions on passenger information, and rigid ticketing and reservation policies require context-aware modeling approaches. Most existing solutions are designed for data-rich environments and may not be directly applicable to such constrained settings.

Based on the above limitations, there is a clear lack of an integrated, context-aware framework that combines advanced spatio-temporal demand forecasting with adaptive and fairness-aware seat allocation mechanisms in railway systems.

### **Proposed Research Contribution**

This research aims to address the identified gap by developing a unified AI-driven framework for intelligent railway capacity management. The proposed approach includes:

1. Constructing high-resolution origin–destination demand matrices using multi-source railway operational data.
2. Benchmarking classical forecasting methods such as ARIMA and Prophet alongside advanced deep learning and graph-based models for spatio-temporal demand prediction.
3. Integrating demand forecasts and uncertainty estimates into adaptive seat allocation strategies using fairness-aware optimization techniques.

4. Evaluating system performance using both predictive accuracy metrics and operational indicators such as seat utilization, passenger equity, and booking rejection rates.

This research is novel in that it integrates spatio-temporal graph-based demand forecasting with fairness-aware adaptive seat allocation within a unified framework. Unlike existing approaches that treat forecasting and allocation as separate processes, the proposed system enables real-time, data-driven decision making by directly linking demand predictions with allocation strategies. Furthermore, the framework is specifically designed to operate under the operational and data constraints of Sri Lanka's railway system, making it both context-aware and practically deployable. By combining advanced AI models with optimization techniques in a single system, this research introduces a creative and application-oriented solution that bridges the gap between theoretical forecasting models and real-world railway operations.

By integrating spatio-temporal demand forecasting with adaptive seat allocation under real-world operational constraints, this research proposes a novel decision-support framework aimed at improving efficiency, fairness, and resource utilization in Sri Lanka's railway transportation system.

## **2. OBJECTIVE**

### **2.1 Main Objective**

To design and implement an AI-driven spatio-temporal demand forecasting and adaptive seat allocation framework for railway transportation systems that achieves improved passenger demand prediction accuracy (MAE reduction of at least 15%) and enhanced seat utilization (increase of at least 20%) compared to existing static allocation methods.

### **2.2 Specific Objectives**

#### **1. Data Collection and Preprocessing**

To collect, integrate, and preprocess multi-source railway data including ticketing records, station-level data, and temporal features in order to construct a structured dataset for modeling, achieving at least 95% data completeness and consistency.

#### **2. Spatio-Temporal Demand Forecasting Model Development**

To develop and train spatio-temporal forecasting models using LSTM and Graph Neural Networks to predict passenger demand across railway stations, achieving improved prediction accuracy with MAE and RMSE compared to baseline models such as ARIMA and Prophet.

#### **3. Adaptive Seat Allocation Optimization**

To design and implement an adaptive seat allocation model using Mixed Integer Linear Programming that dynamically allocates seats based on predicted demand achieving at least 20% improvement in seat utilization compared to existing static allocation approaches.

#### 4. Forecasting Model Performance Evaluation

To evaluate the forecasting models using quantitative metrics such as MAE RMSE and MAPE and validate model performance under both normal and peak demand scenarios.

#### 5. Seat Allocation Model Effectiveness Analysis

To assess the effectiveness of the adaptive seat allocation model using operational metrics such as seat utilization rate passenger demand satisfaction and booking rejection rate achieving measurable reduction in overcrowding and unmet demand.

#### 6. Prototype System Development

To develop and validate a prototype decision-support system that integrates demand forecasting and seat allocation modules and demonstrates its effectiveness through simulation experiments based on real world railway scenarios.

### 3. METHODOLOGY

This research adopts a **Design Science Research (DSR) approach combined with an experimental methodology** to develop and evaluate an AI-driven spatio-temporal demand forecasting and adaptive seat allocation framework for railway transportation systems. The Design Science approach is suitable as the study focuses on designing and implementing an intelligent decision-support artifact, while the experimental approach is used to evaluate its performance under different scenarios.

The overall methodology consists of several stages including problem identification, data collection, data preprocessing, spatio-temporal demand forecasting, adaptive seat allocation optimization, system implementation, and performance evaluation.

#### 3.1 Problem Identification

The first stage focuses on identifying the challenges associated with fluctuating passenger demand and inefficient seat utilization in railway systems. Existing systems rely on static scheduling and rule-based allocation, which fail to adapt to dynamic demand patterns across time and locations.

Based on the literature review, this research addresses the gap by integrating spatio-temporal demand forecasting with adaptive seat allocation within a unified AI-driven framework.

### 3.2 Data Collection

The proposed system requires historical and simulated railway data to train and evaluate the models. Data will be collected from railway reservation systems, historical passenger datasets, and publicly available transportation data sources.

The dataset may include the following attributes:

- Station origin and destination
- Passenger demand counts
- Date and time of travel
- Train schedules
- External factors such as weather conditions and public holidays

These variables help capture both **temporal patterns and spatial relationships** within the transportation network.

To ensure ethical compliance:

- All passenger data will be anonymized to remove personally identifiable information
- Data will be aggregated to prevent individual tracking
- Ethical approval will be obtained where required
- Data usage will comply with privacy and data protection regulations

### 3.3 Data Preprocessing

Before training the forecasting models, the collected datasets will undergo several preprocessing steps to ensure data quality and improve model performance.

The preprocessing process includes:

- **Data Cleaning** – Removing missing values and correcting inconsistent records.
- **Data Normalization** – Scaling numerical features to improve model training.
- **Feature Engineering** – Creating additional features such as day of week, peak hour indicators, and seasonal patterns.
- **Train-Test Splitting** – Dividing the dataset into training and testing sets for model evaluation.

These preprocessing steps are essential for preparing transportation data for machine learning algorithms.

### 3.4 Spatio-Temporal Demand Forecasting

To accurately predict passenger demand, this research employs deep learning models capable of capturing both temporal and spatial dependencies in transportation data.

#### Long Short-Term Memory (LSTM)

Long Short-Term Memory networks are a type of recurrent neural network specifically designed for modeling sequential data. LSTM models can capture long-term temporal dependencies and are widely used for time-series forecasting tasks. Previous studies have demonstrated that LSTM models perform effectively in predicting transportation demand and traffic flow patterns.

In this research, LSTM will be used to learn temporal demand patterns such as daily, weekly, and seasonal variations in passenger demand.

#### Graph Neural Networks (GNN)

Transportation systems can be represented as graph structures where stations act as nodes and railway connections represent edges. Graph Neural Networks are capable of learning spatial relationships within such graph-based structures.

In this study, GNN models will be used to capture spatial dependencies between railway stations, enabling the forecasting model to understand how passenger demand propagates across the transportation network.

By combining LSTM and GNN models, the proposed framework can effectively model spatio-temporal demand patterns.

### 3.5 Adaptive Seat Allocation Optimization

After predicting passenger demand, an adaptive seat allocation mechanism will be developed to distribute available seats efficiently across railway routes.

The seat allocation problem will be formulated as an **optimization problem**. Mixed Integer Linear Programming (MILP) will be used to determine the optimal allocation of seats while satisfying operational constraints such as train capacity and route demand.

The optimization model will aim to:

- maximize seat utilization
- minimize overcrowding
- ensure fair seat distribution across routes

#### Constraints

- Train capacity limits

- Route-specific demand
- Operational policies

### Algorithms

- MILP optimization
- Heuristic methods for large-scale scenarios

This dynamic allocation mechanism allows transportation operators to adapt seat allocation decisions based on predicted demand patterns.

### 3.6 Tools and Platform

The system is implemented using the following tools:

- **Python** – primary programming language for model development
- **TensorFlow / PyTorch** – deep learning frameworks for LSTM and GNN models
- **Scikit-learn** – baseline models and preprocessing
- **NetworkX / PyTorch Geometric** – graph-based modeling for GNN
- **Pandas and NumPy** – data processing and analysis
- **Optimization libraries (PuLP / OR-Tools)** – MILP implementation

These tools are selected due to their scalability flexibility and strong support for machine learning and optimization tasks.

### 3.7 System Evaluation

The performance of the proposed forecasting and optimization framework will be evaluated using both prediction accuracy metrics and operational efficiency metrics.

Forecasting performance will be evaluated using standard metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

These metrics measure the accuracy of predicted passenger demand values.

In addition, seat allocation performance will be evaluated using operational indicators including:

- Seat utilization rate
- Passenger demand satisfaction
- Booking rejection rate
- Reduction in overcrowding

### Validation Approach

- Comparison with baseline models (ARIMA Prophet)
- Scenario-based simulation (peak and off-peak conditions)
- Sensitivity analysis for demand variation

These evaluation metrics will help assess the effectiveness of the proposed system in improving transportation efficiency.

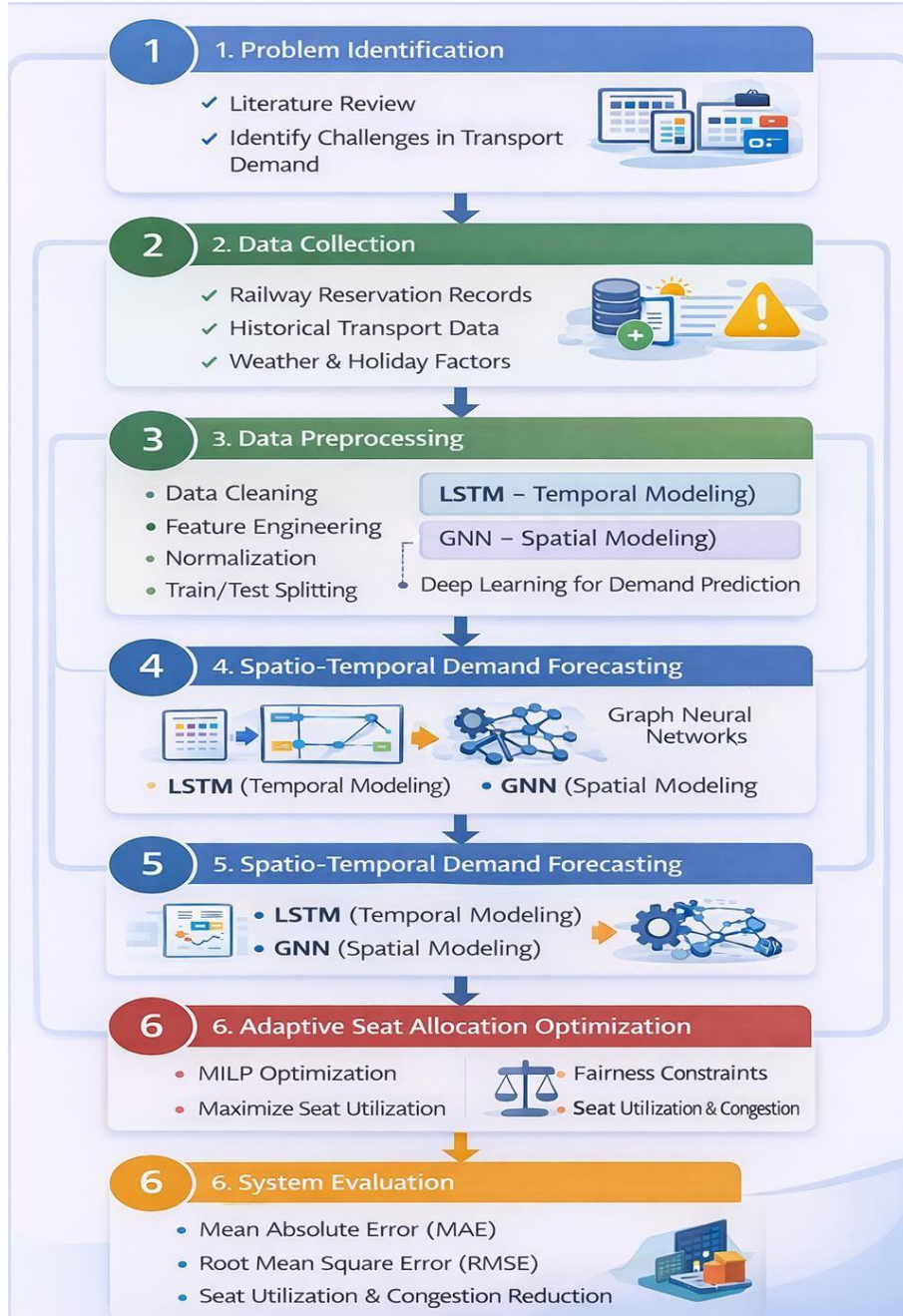


Figure 1: Research Methodology for the Proposed Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation

## 4. HIGH-LEVEL SYSTEM ARCHITECTURE

### 4.1 System Overview

The proposed system is an AI-driven railway decision-support framework designed to improve passenger demand prediction and optimize seat allocation. The system integrates multiple components including data collection, data processing, demand forecasting, optimization, and visualization to enable data-driven decision-making.

The overall architecture consists of five main components:

1. Data Collection Layer
2. Data Processing Layer
3. Spatio-Temporal Demand Forecasting Module
4. Adaptive Seat Allocation Optimization Engine
5. Visualization and Decision Support Dashboard

Among these components, the **Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation modules represent the core contribution of this research.**

### 4.2 System Components

#### 1. Data Collection Layer

This layer is responsible for gathering data from multiple sources including:

- Railway ticketing systems
- Historical passenger demand records
- Train schedules and route data
- External data such as weather and public events

These data sources provide both temporal and spatial information required for demand forecasting. 2.

#### Data Processing Layer

The collected data is cleaned, transformed, and structured in this layer.

Key functions include:

- Data cleaning and missing value handling
- Feature engineering (peak hours seasonal patterns)
- Construction of origin–destination demand matrices
- Data normalization and formatting

This layer ensures that the data is suitable for machine learning models.

#### 3. Spatio-Temporal Demand Forecasting Module (Your Component)

This is the core predictive component of the system.

It consists of:

- Temporal Model (LSTM) → captures time-based demand patterns
- Spatial Model (GNN) → captures relationships between stations
- Hybrid Model → combines both temporal and spatial features

Output:

- Predicted passenger demand for each origin–destination pair over time

This module enables accurate forecasting of passenger demand across railway networks.

#### 4. Adaptive Seat Allocation Optimization Engine (Your Component)

This module uses predicted demand to allocate seats dynamically.

Key parts:

- Demand input from forecasting module
- Optimization model (MILP)
- Constraints (capacity fairness operational rules)

Objectives:

- Maximize seat utilization
- Minimize overcrowding
- Ensure fair distribution of seats

Output:

- Optimized seat allocation plan for trains and routes

#### 5. Visualization Dashboard

This layer provides an interface for railway operators. Features:

- Demand prediction visualization
- Seat allocation recommendations
- Performance metrics and alerts

This supports real-time decision-making.

### 4.3 System Diagram

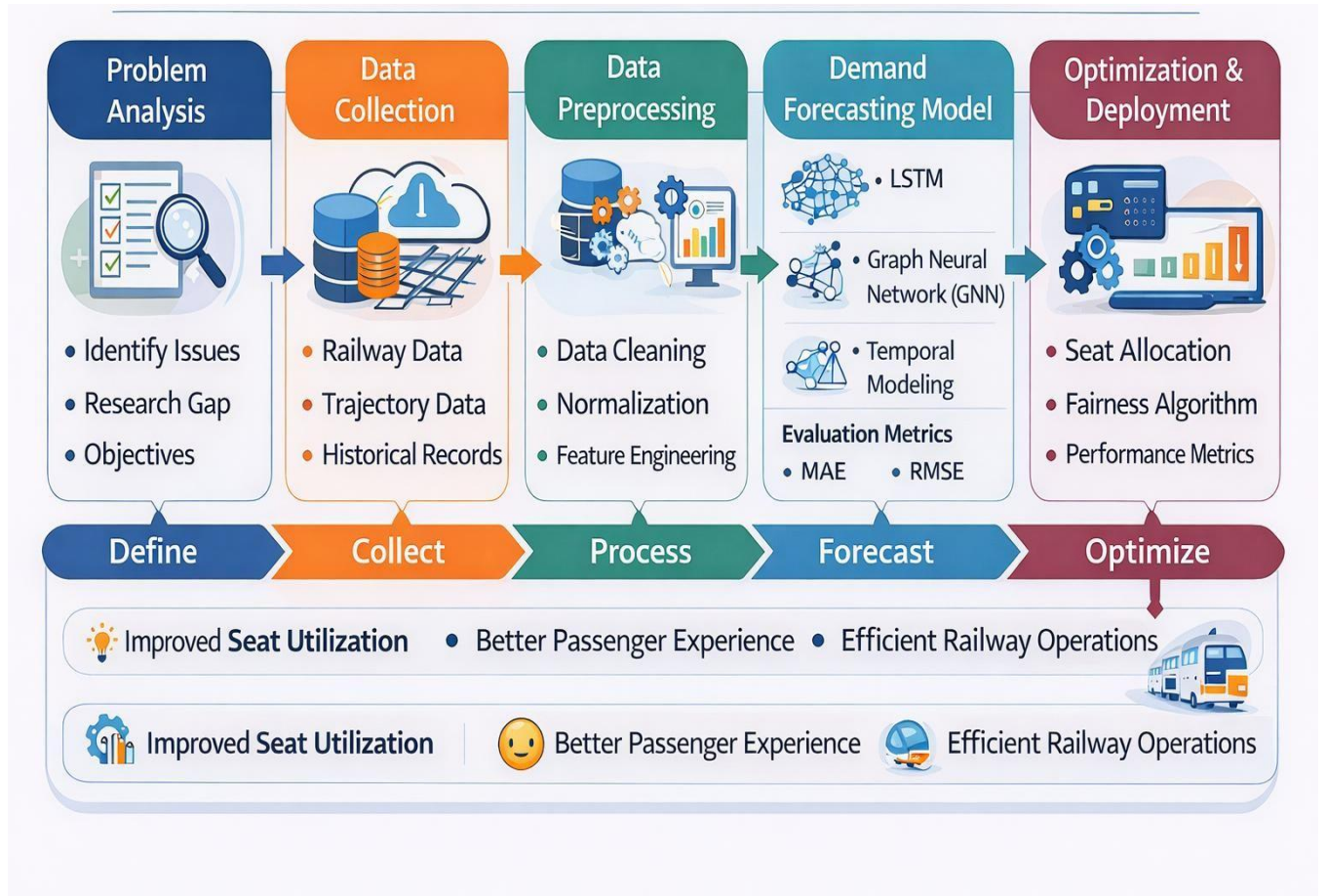


Figure 2: High-Level System Architecture of the Proposed AI-Based Demand Forecasting and Seat Allocation Framework

#### 4.4 Component Interactions

- Data Collection Layer feeds raw data into Data Processing
- Data Processing prepares structured datasets for forecasting
- Forecasting Module predicts demand patterns
- Seat Allocation Engine uses predicted demand for optimization
- Dashboard displays outputs for decision-makers

#### 4.5 Data Flow Explanation

1. Raw data is collected from railway systems and external sources
2. Data is cleaned and transformed into structured format
3. Forecasting models predict passenger demand across stations
4. Predicted demand is passed to the optimization engine
5. Optimization model generates seat allocation plans
6. Results are visualized for operators

#### 4.6 Scalability and Security Considerations

##### Scalability

- System supports large-scale railway networks
- Can handle increasing passenger data over time
- Modular design allows adding new components

## Security

- Passenger data is anonymized
- Secure data storage and access control
- Compliance with data protection regulations

## 5. USER REQUIREMENTS

The user requirements for the proposed system were derived by analyzing the operational needs of railway administrators, transport planners, and passengers. These requirements ensure that the system effectively supports demand forecasting, adaptive seat allocation, and decision-making processes in railway transportation management.

### 5.1 Stakeholder Identification

The proposed system is designed to support multiple stakeholders involved in railway transportation management. These stakeholders interact with the system at different levels to improve decisionmaking, operational efficiency, and passenger experience.

*Table 3: Stakeholders Identification*

| Stakeholder            | Description                                                                | Role in the System                                                  |
|------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------|
| Railway Administrators | Personnel responsible for managing railway operations and scheduling       | Use system outputs for operational planning and resource allocation |
| Transport Planners     | Analysts responsible for transportation policy and infrastructure planning | Use demand forecasts to optimize routes and capacity planning       |
| Passengers             | Individuals who travel using railway services                              | Benefit from improved seat availability and reduced overcrowding    |
| System Administrators  | Technical personnel responsible for maintaining the system                 | Manage system performance, updates and security                     |

### 5.2 Functional Requirements

Functional requirements define the core functionalities that the proposed system must provide to support spatio-temporal demand forecasting and adaptive seat allocation.

*Table 4: Functional Requirements*

| ID | Requirement Description |
|----|-------------------------|
|----|-------------------------|

|      |                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| FR1  | The system shall allow users to upload or input railway operational data such as ticket reservations train schedules and route information |
| FR2  | The system shall preprocess upload datasets by cleaning transforming and preparing the data for model training                             |
| FR3  | The system shall generate temporal features such as time-of-day day of week and seasonal patterns from historical data                     |
| FR4  | The system shall model spatial relationships between railway stations using graph-based techniques                                         |
| FR5  | The system shall generate passenger demand forecasts for origin–destination station pairs using AI models                                  |
| FR6  | The system shall execute the adaptive seat allocation module based on predicted demand                                                     |
| FR7  | The system shall allocate seats while satisfying train capacity and operational constraints                                                |
| FR8  | The system shall ensure fairness in seat allocation across routes and passenger groups                                                     |
| FR9  | The system shall provide visualization of demand forecasts and seat allocation results through dashboards                                  |
| FR10 | The system shall allow administrators to monitor system performance forecasting accuracy and operational metrics                           |
| FR11 | The system shall store and retrieve historical forecasts and allocation results for analysis                                               |

### 5.3 Non-Functional Requirements

Non-functional requirements describe system quality attributes that ensure the system performs efficiently, securely, and reliably.

*Table 5: Non-Functional Requirements*

| Requirement Category | Description                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance          | The system should generate passenger demand forecasts within <b>5 seconds</b> for each route query under normal operating conditions.           |
| Scalability          | he system shall protect passenger and operational data through encryption authentication and access control mechanisms                          |
| Reliability          | The forecasting models should maintain consistent prediction performance and system availability during operation with minimal system failures. |
| Security             | Passenger and operational data must be protected through secure data storage, authentication mechanisms, and controlled access permissions.     |

|                 |                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usability       | The system shall provide an intuitive dashboard with clear visualizations for easy interpretation by users                                                                           |
| Maintainability | The system architecture should support easy updates, retraining, and modification of forecasting models and optimization algorithms without major system redesign.                   |
| Availability    | The system should maintain high availability, ensuring that authorized users can access forecasting and seat allocation services 24/7 with minimal downtime during normal operation. |

## 6. COMMERCIALIZATION PLAN

The commercialization plan outlines how the proposed AI-driven spatio-temporal demand forecasting and adaptive seat allocation system can be deployed and adopted by railway authorities and transportation organizations to improve operational efficiency and passenger experience.

### 6.1 Target Market and Customer Persona

The primary target market includes organizations responsible for managing railway and public transportation systems.

#### Target Market

- Sri Lanka Railway Department
- Urban transportation authorities
- Metro and rail operators
- Transportation planning agencies
- Smart city development initiatives

#### Customer Persona

The primary users of the system are:

- **Railway Operations Manager**  
Responsible for scheduling and resource allocation  
Needs accurate demand forecasts and optimized seat allocation
- **Transport Planner / Data Analyst**  
Responsible for planning infrastructure and services  
Needs data-driven insights for decision-making
- **System Administrator**  
Responsible for system maintenance and performance

Needs a secure scalable and reliable system

These users require intelligent tools to improve passenger demand management and operational efficiency.

## 6.2 Value Proposition

The proposed system provides several benefits that distinguish it from existing transportation management tools.

Key value propositions include:

- Accurate passenger demand forecasting using AI models
- Dynamic seat allocation based on predicted demand patterns
- Improved railway capacity utilization
- Reduced passenger overcrowding and waiting time
- Data-driven decision support for transportation planning

This integrated approach differentiates the system from existing solutions that treat forecasting and allocation separately.

## 6.3 Revenue Model

The proposed system could generate revenue through several potential business models.

### Software Licensing

Transportation authorities can purchase a license to deploy the system within their existing railway management infrastructure.

### Software-as-a-Service (SaaS)

The system can be provided as a cloud-based service where railway authorities pay a subscription fee for access to forecasting and analytics tools.

### Consulting and Analytics Services

Transportation agencies may require customized analytics solutions for transportation planning, which can generate additional consulting revenue.

## 6.4 Cost Estimation

The estimated costs for developing and maintaining the system include:

- **Subscription-based pricing (SaaS):** Monthly or annual plans based on system usage
- **Tiered pricing model:** Different pricing levels based on number of stations routes and data volume
- **Enterprise licensing:** Custom pricing for large railway authorities

- **Pilot deployment:** Low-cost or trial-based implementation to encourage adoption

## 6.5 Pricing Strategy

The pricing strategy is designed to ensure affordability and scalability:

- **Subscription-based pricing (SaaS):** Monthly or annual plans based on system usage
- **Tiered pricing model:** Different pricing levels based on number of stations routes and data volume
- **Enterprise licensing:** Custom pricing for large railway authorities
- **Pilot deployment:** Low-cost or trial-based implementation to encourage adoption

This flexible pricing model supports both small-scale and large-scale transportation systems.

## 6.6 Competitive Advantage

The proposed system offers several advantages compared to existing transportation management systems.

- Integration of **spatio-temporal forecasting and seat allocation** in a single platform
- Use of **advanced AI techniques such as LSTM and Graph Neural Networks**
- Fairness-aware optimization strategies for seat allocation
- Scalable architecture suitable for smart transportation systems

These features position the system as a valuable tool for improving railway capacity planning and operational efficiency.

## 6.7 Intellectual Property Considerations

The system includes several innovative components that can be protected as intellectual property:

- AI-based spatio-temporal forecasting model architecture
- Integrated forecasting and seat allocation framework
- Optimization algorithms and system design

Potential IP strategies include:

- Copyright protection for software and code
- Patent opportunities for novel algorithms and system integration methods
- Licensing agreements for commercial deployment

Protecting intellectual property ensures long-term commercial value and competitive advantage.

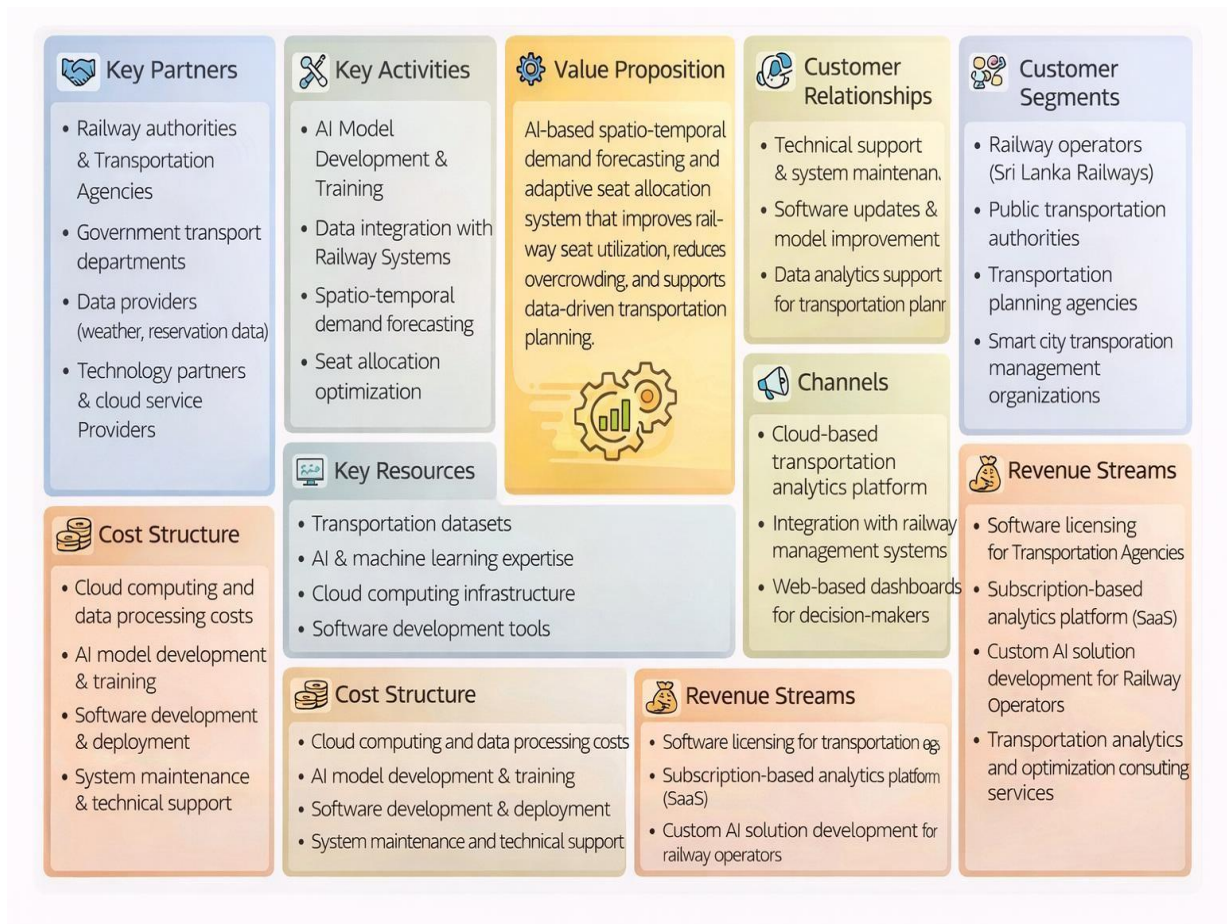


Figure 3: Business Model Canvas for the Proposed AI-Based Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation System

## 7. BUDGET AND JUSTIFICATION

The proposed research requires computational resources, software tools, data storage, and human effort to develop and evaluate the AI-based spatio-temporal demand forecasting and adaptive seat allocation system. The estimated budget is summarized in Table 6.

Table 6: Budget Estimation

| Item                                | Description                                                                                | Estimated Cost (LKR) |
|-------------------------------------|--------------------------------------------------------------------------------------------|----------------------|
| <b>Cloud Computing Services</b>     | GPU-based cloud resources for training LSTM and GNN models                                 | 45,000               |
| <b>Data Storage</b>                 | Secure cloud and local storage for datasets and model outputs                              | 10,000               |
| <b>Software Tools and Libraries</b> | Development tools and libraries (most open-source with minor subscription costs if needed) | 15,000               |
| <b>External Storage Device</b>      | Backup storage for datasets models and project files                                       | 12,000               |

|                                 |                                                                        |        |
|---------------------------------|------------------------------------------------------------------------|--------|
| <b>Research Materials</b>       | Printing documentation and report preparation                          | 8,000  |
| <b>Internet and Data Access</b> | High-speed internet for data collection and research activities        | 10,000 |
| <b>Human Effort Cost</b>        | Effort for system development model training testing and documentation | 20,000 |
| <b>Contingency</b>              | Unexpected expenses during project development                         | 10,000 |

Total Estimated Cost: LKR 130,000

### **Budget Justification**

The budget allocation is designed to ensure that the research project can be conducted efficiently while maintaining reasonable costs.

### **Cloud Computing Resources**

Deep learning models such as LSTM and Graph Neural Networks require high computational power for training and experimentation. Cloud-based GPU services enable faster model training and scalability compared to local systems.

### **Data Storage**

Secure storage is required to manage large transportation datasets trained models and experimental outputs. Both cloud and local storage solutions ensure data availability and reliability.

### **Software Tools**

The system is developed using tools such as Python TensorFlow PyTorch and data analytics libraries. Most tools are open source but additional costs may arise from premium features or development environments.

### **External Storage Device (Hardware)**

A portable storage device is required to back up datasets trained models and project files ensuring data safety and recovery in case of system failure.

### **Research Materials**

Printing costs, documentation preparation, and report binding are required for proposal submission and final dissertation submission. **Internet and Data Access**

Reliable internet access is required to download research papers, datasets, and software libraries used for developing the proposed system.

## Human Effort Cost

This includes the effort required for data preprocessing model development system implementation testing and documentation. It reflects the time and expertise involved in building the AI-based system.

## Contingency

A contingency allocation is included to address unexpected costs that may arise during project development.

## 8. Risk Analysis

Risk analysis is an essential component of research project planning as it helps identify potential challenges that may affect the successful development and implementation of the proposed system. In this study, several technical, data-related, and operational risks associated with spatio-temporal demand forecasting and adaptive seat allocation are identified and evaluated. Each risk is assessed based on its likelihood of occurrence and its potential impact on system performance. Appropriate mitigation strategies are defined to minimize the negative effects of these risks.

Table 7: Risk Analysis

| Risk                                                                   | Risk Level | Mitigation Strategy                                                                                                               |
|------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Limited access to railway operational datasets                         | High       | Use publicly available transportation datasets and generate realistic simulated datasets to support model training and validation |
| Model overfitting during training of AI models                         | Medium     | Use cross-validation regularization techniques and hyperparameter tuning to improve generalization performance                    |
| Insufficient computational resources for training deep learning models | Medium     | Utilize cloud computing platforms such as Google Colab AWS or GPU-based services for scalable model training                      |
| Data privacy and confidentiality concerns                              | High       | Use anonymized datasets and follow ethical data handling practices to ensure compliance with data protection regulations          |
| System integration challenges with existing railway infrastructure     | Medium     | Adopt modular architecture and perform incremental integration and testing to ensure compatibility                                |
| Data quality issues such as missing or inconsistent data               | High       | Apply data preprocessing techniques including cleaning normalization and data validation to improve data quality                  |

|                                                             |        |                                                                                                                               |
|-------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------|
| Forecasting model accuracy not meeting expected performance | Medium | Compare multiple models including LSTM GNN ARIMA and Prophet and select the best-performing model based on evaluation metrics |
| Optimization model scalability issues for large datasets    | Low    | Use heuristic optimization techniques or approximate solutions for large-scale scenarios                                      |

## 9. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) defines the tasks required to develop the spatio-temporal demand forecasting and adaptive seat allocation component. The work is organized into phases and aligned with a timeline to ensure balanced workload distribution and efficient progress.

### 9.1 Task Breakdown and Timeline

Table 8: Task breakdown and timeline

| Task ID | Task Description                                            | Duration | Timeline (Weeks) |
|---------|-------------------------------------------------------------|----------|------------------|
| T1      | Literature review on demand forecasting and seat allocation | 2 weeks  | Week 1 – 2       |
| T2      | Data collection and dataset preparation                     | 2 weeks  | Week 3 – 4       |
| T3      | Data preprocessing and feature engineering                  | 2 weeks  | Week 5 – 6       |
| T4      | Development of baseline models (ARIMA Prophet)              | 1 week   | Week 7           |
| T5      | Development of LSTM model (temporal forecasting)            | 2 weeks  | Week 8 – 9       |
| T6      | Development of GNN model (spatial modeling)                 | 2 weeks  | Week 10 – 11     |
| T7      | Integration of spatio-temporal forecasting model            | 1 week   | Week 12          |
| T8      | Development of seat allocation optimization model (MILP)    | 2 weeks  | Week 13 – 14     |
| T9      | Integration of forecasting and allocation modules           | 1 week   | Week 15          |
| T10     | System testing and evaluation                               | 2 weeks  | Week 16 – 17     |
| T11     | Performance analysis and result validation                  | 1 week   | Week 18          |
| T12     | Documentation and report writing                            | 2 weeks  | Week 19 – 20     |

## 8.2 Task Breakdown

### 1. Demand Forecasting Tasks

- Data preprocessing and feature engineering
- LSTM model development

- GNN model development
- Spatio-temporal model integration
- Model evaluation

## **2. Seat Allocation Tasks**

- Problem formulation (MILP)
- Constraint definition (capacity fairness)
- Optimization model implementation
- Result validation

## **3. Integration Tasks**

- Connecting forecasting output to allocation module
- System-level testing
- Performance evaluation

### **8.3 Balanced Workload Distribution**

The workload is distributed across the development phases to ensure efficiency:

- Early Phase (Weeks 1–6): Research and data preparation
- Middle Phase (Weeks 7–14): Model development and optimization
- Final Phase (Weeks 15–20): Integration testing and documentation

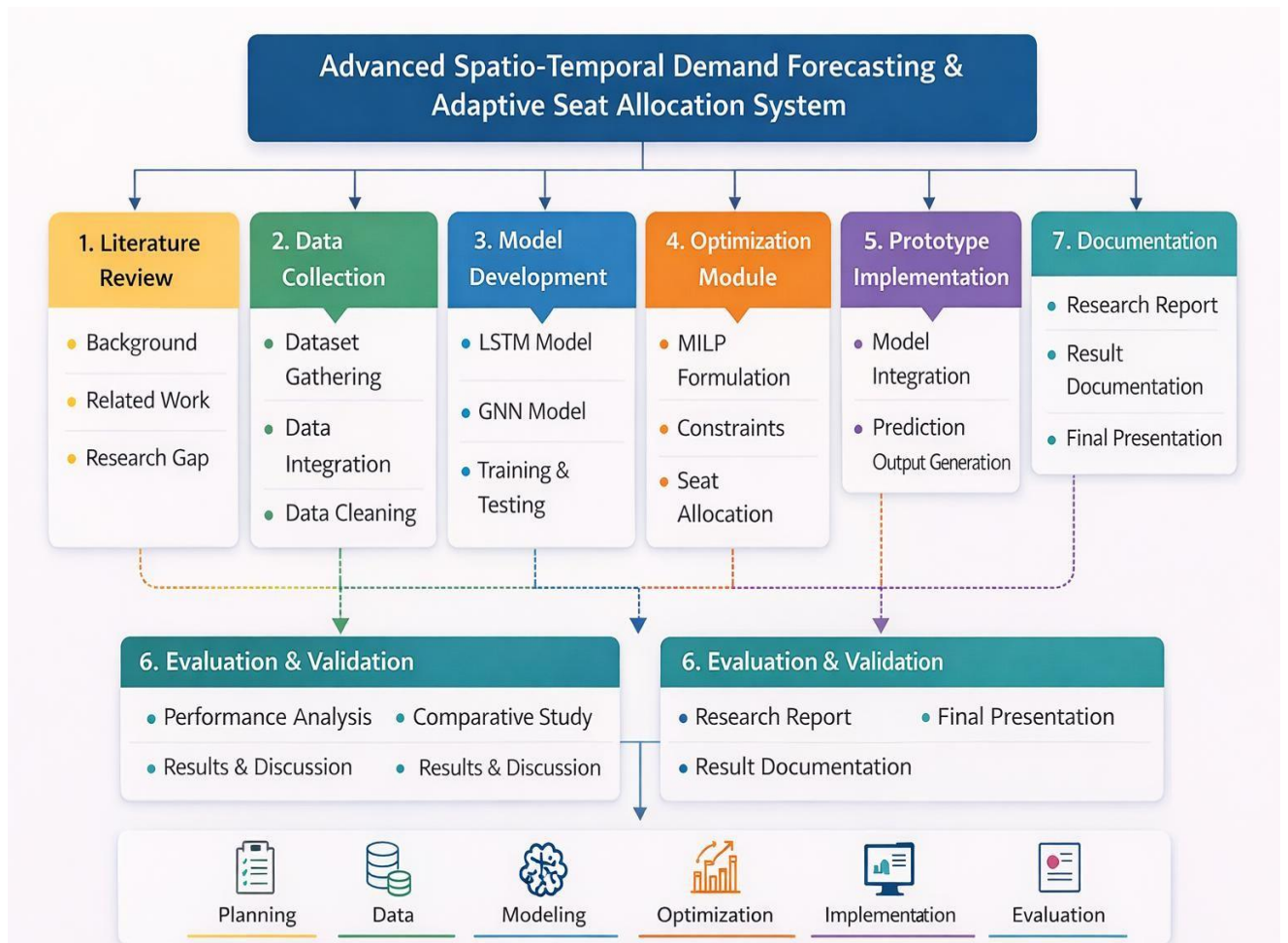
Each phase includes a balanced mix of analytical and implementation tasks to avoid workload concentration in a single period.

### **8.4 Timeline Alignment**

The WBS is aligned with the overall research timeline:

- Sequential dependency between tasks (data → model → optimization → evaluation)
- Parallel development of forecasting and optimization components where possible •  
Sufficient time allocated for testing and validation

This structured approach ensures timely completion of the component within the project duration.



**Figure X:** Work Breakdown Structure for the Proposed Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation System

*Figure 4: Work Breakdown Structure (WBS) for the Proposed Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation System*

## 10. GANTT CHART

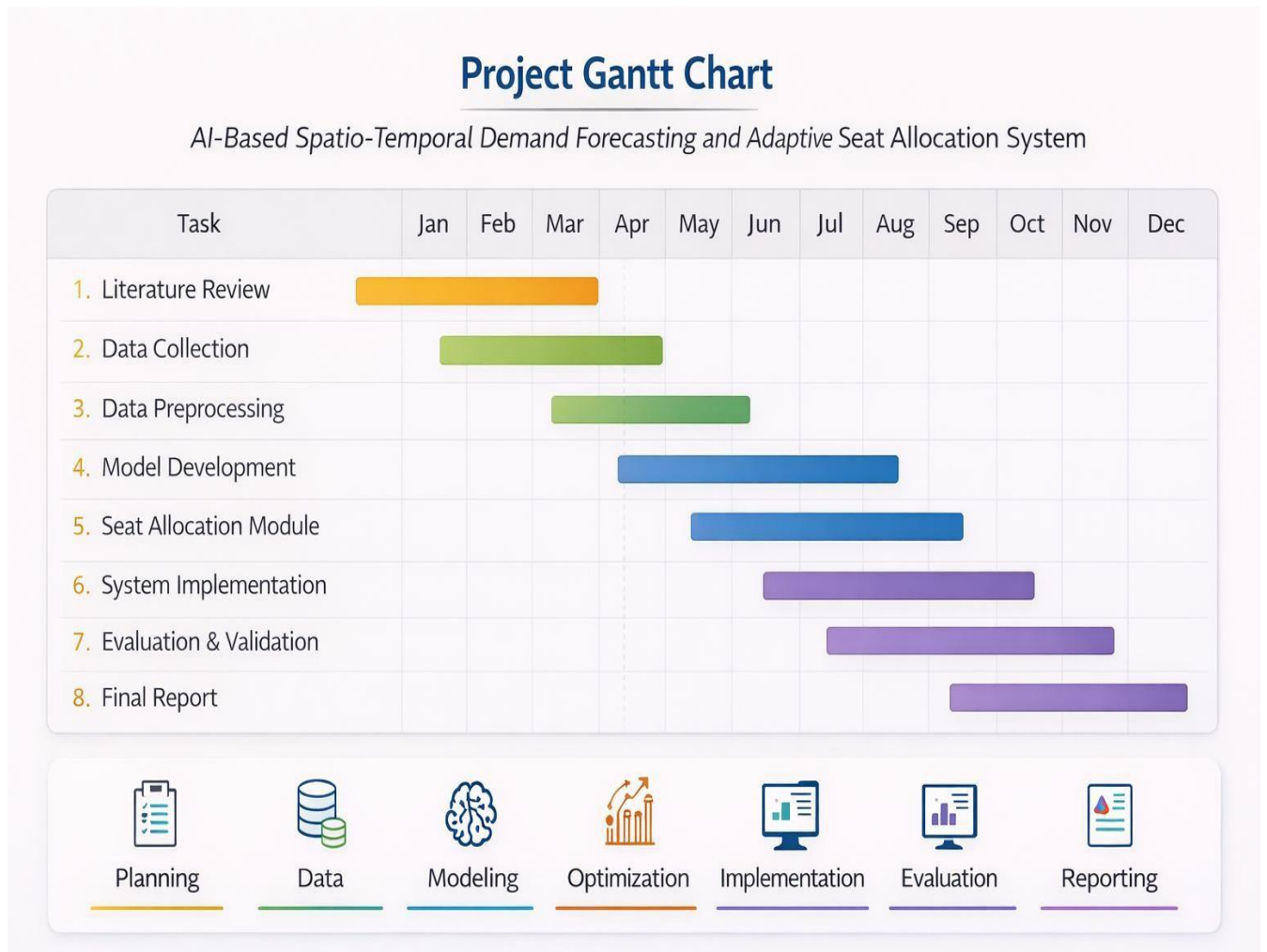


Figure 5: Project Gantt Chart for the Proposed Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation System

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## 12. APPENDICES

### APPENDIX A – Simple Dataset Structure

The dataset used for the proposed system consists of both spatial and temporal attributes related to railway passenger demand.

*Table 9: Sample Dataset Structure*

| Attribute           | Description                           |
|---------------------|---------------------------------------|
| Origin Station      | Starting station of passenger journey |
| Destination Station | Ending station of passenger journey   |
| Date                | Travel date                           |
| Time                | Travel time                           |
| Passenger Count     | Number of passengers                  |
| Train ID            | Train identifier                      |
| Weather Condition   | External factors influencing demand   |
| Day Type            | Weekday or weekend indicator          |

This dataset is used to construct origin–destination demand matrices for model training.

### APPENDIX B – System Architecture Diagram

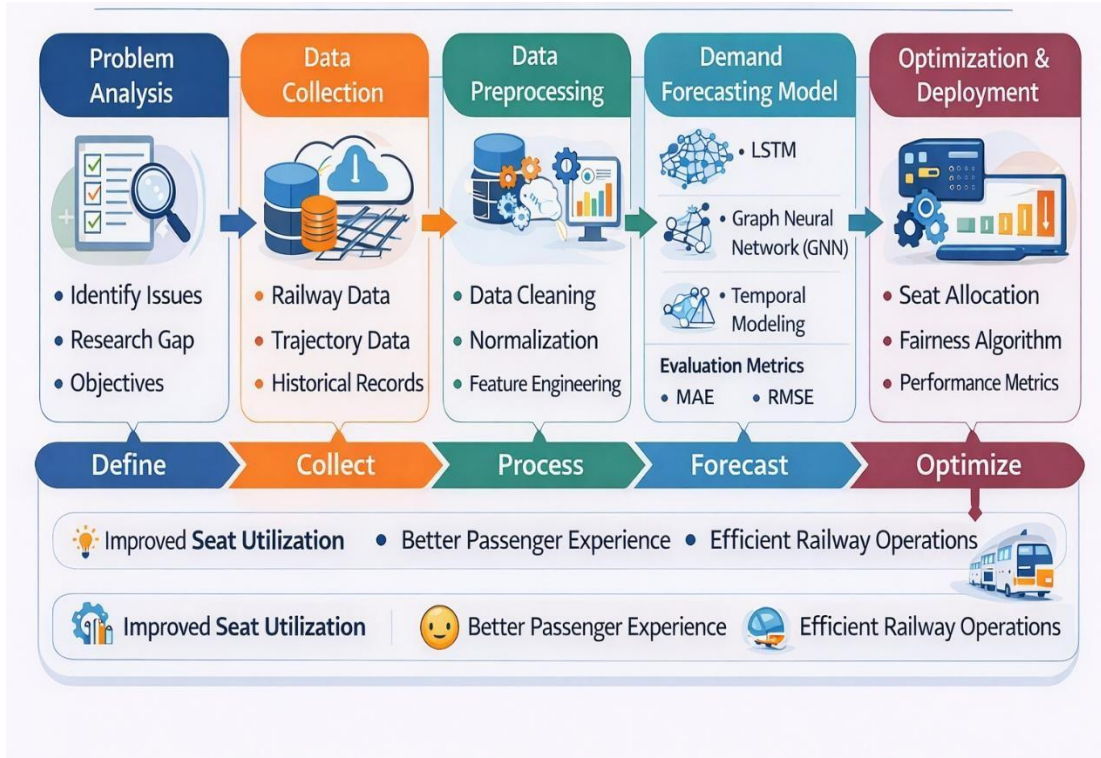


Figure 2 illustrates the overall system architecture. Data is collected from railway systems and processed before being passed to the spatio-temporal forecasting module. The predicted demand is then used by the adaptive seat allocation module to optimize seat distribution. The results are visualized through a dashboard for decision-making.

## APPENDIX C – Model Architecture

### C1: LSTM Model (Temporal Modeling)

The LSTM model is used to capture time-based demand patterns such as daily and seasonal variations.

**Input:** Time-series passenger data

**Output:** Predicted passenger demand

### C2: Graph Neural Network (Spatial Modeling)

The GNN model represents railway stations as nodes and routes as edges to capture spatial dependencies.

**Input:** Railway network graph

**Output:** Spatial demand relationships

### C3: Spatio-Temporal Model Integration

The outputs of LSTM and GNN models are combined to generate accurate demand predictions across both time and space.

## **APPENDIX D – Ethical Considerations**

The proposed system follows ethical guidelines in handling data:

- Passenger data is anonymized
- No personally identifiable information is used
- Data is used only for research purposes
- Secure storage and controlled access are maintained